

Answer Position and Lexical Pruning Shape Synthetic Clause Retrieval in a Small Language Model

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Abstract

Does adding context hurt a small language model, and do the location and composition of that context matter? We evaluate the frozen Qwen2.5-0.5B-Instruct Q8 artifact on 180 synthetic contract-clause probes. Deterministic runs span mean prompt lengths of 1989, 3986, and 7992 tokens. Moving the answer clause from the start to the end raised accuracy by 58.3 points at the shortest tier and 50.0 points at the middle tier. Keyword pruning changed matched accuracy by -33.3 points, whereas selecting sentences that named the question’s company reached 100.0% accuracy and improved on keyword pruning by 93.3 points. The matched comparison across prompt lengths was inconclusive, and the filler manipulation does not isolate topical relevance.

1. Introduction

You paste a long contract into a chatbot and ask, “What code renews the vendor agreement?” The answer sits in one clause, but a similar buyer clause repeats the words in your question and gives a different code.

This ordinary lookup becomes less predictable in a long prompt. The model may fail because it must process a longer document, because the answer appears in a weak position, or because nearby text resembles the question. Each explanation suggests a different response: shorten the prompt, move the key clause, or filter the document. Distinguishing among them requires an evaluation that varies a single property while holding the others fixed.

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We study these choices in a synthetic, gradeable version of the contract situation. Each document contains an answer-bearing vendor clause and a look-alike clause for another company. The question uses the wording of the look-alike clause, while the answer clause expresses the same relation with different words. This construction reduces the value of literal matching, following the motivation of NoLiMa (Modarressi et al., 2025).

We provide a paired evaluation of a frozen open model at laptop scale. Prior work established position and length failures in larger systems (Liu et al., 2023; Levy et al., 2024) and showed that irrelevant statements can disrupt reasoning (Shi et al., 2023). Our comparisons match prompt lengths, answer positions, and questions while varying whether the filler is code-heavy business text or unrelated prose. We compare a deterministic keyword filter with a query-entity selector that keeps sentences naming the company in the question.

The experiment asks four questions:

1. As mean prompt length grows from 1989 to 7992 tokens, how much does accuracy change on shared cases?
2. How does moving the same answer clause among the start, middle, and end change accuracy?
3. How does accuracy differ between code-heavy business filler and unrelated prose in matched documents?
4. Which cheap selector better preserves accuracy: keyword overlap or matching the company named in the question?

The experiment does not identify a general context-length threshold. Within this grid, accuracy was higher when the answer appeared near the end and lower with code-heavy business filler than with unrelated prose. Keyword pruning amplified the lexical trap, while matching the query entity avoided it.

2. Related Work

Liu et al. (2023) found a U-shaped position curve in multi-document question answering and key-value retrieval, with stronger use of information near context boundaries. We conduct a smaller clause-reordering test in which the question, filler sequence, clauses, and codes remain fixed while we insert the answer clause at a different depth. When we place it before the competing clause, the competing clause’s final sentence index shifts by one. The resulting pattern mainly favors recent information rather than forming a full U-shape, because accuracy is also low in the start condition.

Prompt length can matter even when the underlying problem is fixed. Levy et al. (2024) padded matched reasoning problems and observed degradation well before model context limits, while RULER found that usable context lengths can be shorter than advertised windows (Hsieh et al., 2024). Our prompt-length comparison includes only cases present at every tier, so the mix of scenarios remains fixed as prompts grow.

Shi et al. (2023) added irrelevant statements to arithmetic problems and found lower reasoning accuracy. We compare filler from the same business domain with unrelated prose while matching prompt length and question wording. The business condition also contains more entities and code-shaped candidates, so the comparison measures that bundle rather than topical relevance alone. NoLiMa shows that synthetic retrieval tests based on literal word matches can overstate retrieval ability (Modarressi et al., 2025). We therefore paraphrase the question in the answer clause while giving the competing clause the same surface wording as the question. The tested model belongs to the instruction-tuned Qwen family (Yang et al., 2024).

3. Probe Design

Model and execution. We evaluate the frozen qwen2.5-0.5b-instruct-q8_0.gguf artifact, whose SHA checksum appears below:

```
ca59ca7f13d0e15a8cfa77bd17e65d24f6844b554a7b6c12e07a5f89f76844e
```

We run it through llama-completion with greedy decoding, zero temperature, and a fixed seed. The complete manifest contains 180 probes, each stored as one prompt and one raw output. Existing outputs are skipped on rerun, so interrupted inference resumes by stable probe identifier.

Scenario families. For each family, seeded code generates the target company and code, a competing company and code, a question, and filler. The question

asks for the renewal code needed to extend the target company’s contract. Its wording exactly matches the competing clause, whereas the answer-bearing clause calls the same value a renewal identifier. A matched literal control reverses the wording of the two clauses while retaining the same filler.

For each full-context family, we insert the answer clause near the start, middle, or end. Family generation does not use target position in any random seed. We assign the competing clause a fractional insertion index, either earlier or later in the document, that is independent of answer position and fixed across the three versions. Its final sentence index shifts by one when the answer is inserted before it. Thus, each paired position contrast changes both the answer placement and this mechanical one-sentence shift while preserving the scenario content and filler order.

Amount and filler composition. The tokenizer, rather than a character estimate, determines prompt length. The three full-context tiers average 1989, 3986, and 7992 tokens. Each tier balances business filler with unrelated sentences. Business filler consists of other companies, records, and unrelated codes; unrelated filler describes places and events outside the contract domain. Filler pairs share the family, target position, question, and measured tier.

This manipulation changes several properties together. Compared with unrelated prose, the related filler contains more business entities, contract-like templates, and code-shaped answer candidates. The paired comparison therefore identifies the effect of these two generated filler conditions, not the effect of topical relevance alone.

The final tier contains fewer families because it is more expensive to run. Prompt-length estimates therefore use only the 12 cases shared by every tier rather than the unequal full-tier totals. Filler-composition estimates use 18, 15, and 6 pairs at the three tiers, respectively.

Selectors and controls. At the middle tier, a keyword filter retains sentences with at least two content words shared with the question, falling back to the highest-overlap sentences if too few qualify. A second selector extracts the company named after “contract with” in the question and keeps sentences containing that name. If extraction or matching fails, it falls back to the keyword filter. Both rules use only visible text, preserve original order, and require no answer labels. The comparisons are restricted to the 30 cases with full context and both selector outputs. We also include

6 closed-book questions and 6 matched literal-versus-paraphrase controls.

The keyword filter retained the answer clause and lexical decoy in all 30 matched prompts. The query-entity selector retained the answer clause in 30 prompts and the decoy in 0. A paired diagnostic removes only the decoy from each keyword-pruned prompt, leaving every other selected sentence fixed. This diagnostic uses the known synthetic label; both automatic selectors do not.

4. Scoring and Comparisons

Normalized accuracy requires the target code to appear without the competing code. Exact-output compliance requires a response containing one code and no other text. A distractor capture contains the competing code without the target, and an invalid response contains both expected codes or neither. We strip only the runner’s fixed transport trailer before applying these rules.

The inferential target is the population of synthetic scenario families defined by the seeded generator. To compute each 95% interval, we resample whole observed families while retaining all positions, filler types, and variants from every selected family. Exact enumeration makes this cluster bootstrap deterministic. Paired changes are differences in accuracy over an exact intersection of scenario keys, and their intervals use the same family-level resampling. The intervals describe sensitivity to the small sample of families, while wins, losses, and ties expose the underlying paired outcomes. Because greedy decoding is deterministic, the intervals do not measure uncertainty from repeated decoding.

5. Results

Amount. The relationship between prompt length and accuracy remains unresolved. Accuracy was 50.0% at a mean prompt length of 1989 tokens and 58.3% at 3986 tokens. This paired change of 8.3 points has a family-cluster interval of [-14.8, 31.5] over 12 cases. At 7992 tokens, accuracy was 58.3%, with 2 wins, 2 losses, and 8 ties relative to the middle tier (Table 1). Because the longest-tier intersection represents only 12 probes from 2 scenario families, these observations neither establish a benefit nor rule out degradation within the measured range.

Position. Moving the answer clause to the end produced the largest and most consistent gain. At the shortest tier, accuracy rose from 16.7% at the start to 75.0% at the end. The paired increase was 58.3

Table 1. Matched amount and filler-composition comparisons. Brackets give 95% family-cluster bootstrap intervals; n is the matched denominator. Changes are later minus earlier for amount and business filler minus unrelated prose for composition.

Comparison	Accuracy A	Accuracy B	Change (n)
Amount: short to middle	50.0 [50.0, 50.0]	58.3 [35.2, 81.5]	8.3 (12)
Amount: middle to long	58.3 [35.2, 81.5]	58.3 [50.6, 66.0]	0.0 (12)
Composition at short tier	44.4 [22.2, 61.1]	22.2 [11.1, 33.3]	-22.2 (18)
Composition at middle tier	46.7 [6.7, 86.7]	33.3 [13.3, 53.3]	-13.3 (15)

Table 2. Accuracy by answer-clause position. Each row compares the same scenario families across positions. Brackets give 95% family-cluster bootstrap intervals.

Mean tokens	Start	Middle	End
1989	16.7 [0.0, 33.3]	8.3 [0.0, 25.0]	75.0 [58.3, 91.7]
3986	20.0 [0.0, 40.0]	30.0 [0.0, 70.0]	70.0 [30.0, 100.0]
7992	25.0 [1.9, 48.1]	75.0 [51.9, 98.1]	75.0 [51.9, 98.1]

points, with a family-cluster interval of [50.0, 75.0], 7 wins, and no losses. At the middle tier, the same move raised accuracy by 50.0 points [20.0, 80.0], with 5 wins and no losses. The sparse longest tier also favored the end over the start by 50.0 points, but it contains only 4 pairs. Accuracy with the clause in the middle was lower than accuracy with it at the end in the two populated tiers (Table 2).

Filler composition. Code-heavy business filler produced lower accuracy than unrelated prose. The matched difference was -22.2 points [-33.3, 0.0] over 18 pairs at the shortest tier and -13.3 points [-46.7, 13.3] over 15 pairs at the middle tier. At the shortest tier, the paired outcomes were 1 win, 5 losses, and 12 ties. At the middle tier, they were 1 win, 3 losses, and 11 ties. The longest-tier estimate was -16.7 points, but its 2 families are too sparse to strengthen that conclusion. Because the business filler also adds code-shaped candidates and repeated entity templates, these results do not show that topical relevance caused the difference.

Mitigation. The query-entity selector reversed the keyword filter’s failure. Full-context accuracy was 40.0%, compared with 6.7% after keyword filtering and 100.0% after query-entity selection. Query-entity selection improved on full context by 60.0 points [33.3, 86.7], with 18 wins and no losses. Compared with keyword pruning, it gained 93.3 points [80.0, 100.0], with 28 wins, no losses, and 2 ties (Table 3).

Removing only the retained decoy raised keyword-pruned accuracy to 90.0%, a change of 83.3 points with 25 wins and no losses. Literal wording changed matched accuracy by 66.7 points, with 4 wins and no losses. Across all outputs, distractor capture was

Table 3. Matched selector results and diagnostic controls. Accuracy intervals are 95% family-cluster bootstrap intervals.

Condition	Accuracy	Count
Full context	40.0 [13.3, 66.7]	30
Keyword pruned	6.7 [0.0, 20.0]	30
Query entity	100.0 [100.0, 100.0]	30
Decoy removed	90.0 [70.0, 100.0]	30
Literal wording	83.3 [50.0, 100.0]	6
Paraphrase wording	16.7 [0.0, 50.0]	6
Closed book	0.0 [0.0, 0.0]	6

40.0%, exact-output compliance was 96.7%, and invalid output was 7.2%. The diagnostic and deployable selector agree that retaining the lexical competitor, rather than deleting the answer, caused the keyword filter’s failure in this grid.

6. Limitations

The findings concern one small quantized model and may not transfer to larger models or other quantization levels. Synthetic clause lookup is not natural contract analysis, and retrieval accuracy is not a measure of general capability. The 45-minute CPU budget capped the grid, leaving only 12 full-context probes at the longest tier and wide intervals throughout. That sparse tier prevents a conclusion about degradation within the measured range, and the experiment cannot address longer prompts. The filler comparison confounds topical relation with semantic domain, entity and code density, candidate count, and sentence templates. A factorial design that matches those properties would be needed to identify a topical-relevance effect. Greedy decoding removes run-to-run sampling variation but does not show how sampling would change the estimates. Query-entity selection assumes that the question names one company exactly and that the relevant sentence repeats that name. It does not address aliases, pronouns, or queries without a unique named subject. The decoy-removal control also depends on an oracle label unavailable in natural documents.

7. Reproducibility

Analysis scripts generate every reported quantity from the committed probe manifest and raw model outputs. This procedure follows the principle that reports of computational evidence should expose their budget and variation (Dodge et al., 2019). The model and cached data are frozen and checksummed, and each raw output retains its stable probe identifier. Run-

ning make all regenerates results.json, the LaTeX value macros, and the paper from cached outputs; running make clean-deep && make all repeats deterministic model inference. Repository gates check frozen inputs, citation resolution, number provenance, pipeline freshness, result sanity, prose style, anonymity, and compilation.

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